Freeriding Entanglement Distillation to Achieve Quantum Network Tomography

***Abstract*—Entanglement distillation and quantum network tomography (QNT) address complementary needs in noisy quantum networks: the former improves the fidelity of distributed entanglement, while the latter identifies which internal links introduce the noise. Both require local measurement, yet they are typically treated as separate tasks. In this paper, we integrate QNT into end-to-end entanglement distillation by reusing the measurement statistics of the double-selection protocol, collected at peripheral monitors, for link-level characterization, so that distillation measurements come "for free" with respect to tomography. For a three-link star network with unknown Pauli channels, we combine Unicast and modified Mergecast preparations with Clifford dressing to identify all three Pauli channels. On this basis, we develop two estimators: the Joint-Coincidence Estimator (JCE), which exploits joint X/Z coincidence statistics under a source-state fidelity condition $F > 1/2$, and the Marginal-Coincidence Estimator (MCE), which uses only Z-basis coincidence statistics with uniform Clifford dressing and removes this condition. We prove that both estimators uniquely identify all three Pauli channels without access to measurement outcomes at the central node. Qiskit simulations show that JCE and MCE recover channel parameters accurately while retaining fidelity-improved entangled pairs, and that retaining joint coincidence information improves estimation accuracy. We further extend JCE to a six-link triangular network, demonstrating applicability to more general network topologies beyond the star topology. These results demonstrate how distillation measurements can support internal-link characterization without additional measurements dedicated to tomography.111**

***Index Terms*—Quantum Network Tomography, Entanglement Distillation, Mergecast.**

I. Introduction

Distributing reliable entanglement across noisy quantum networks is essential for secure communication, distributed quantum computing, and networked quantum sensing [1]. Shared entanglement enables quantum information processing between distant nodes, but its usefulness is limited by noise introduced during distribution. Entanglement distillation and quantum network tomography (QNT) address two complementary aspects of this problem.

Entanglement distillation uses local operations and classical communication (LOCC) to probabilistically convert multiple noisy entangled pairs into fewer pairs of higher fidelity [2, 3]. It provides a means to recover useful entanglement from imperfect physical links and is an important component of quantum repeater architectures [4]. For distributed quantum computing, improving shared entanglement supports more reliable teleportation and quantum operations between remote processors [1].

QNT infers the noise parameters of internal quantum links from measurements performed at accessible peripheral nodes, termed *monitors* [5]. For experimental quantum networks, these estimates help diagnose noisy links and guide distillation protocols. For distributed quantum computing, they provide information for predicting communication errors and selecting entanglement distribution paths. The fidelity of an end-to-end state alone does not identify which internal links introduce the noise; QNT seeks to recover this link-level information from endpoint observations. Together, entanglement distillation and QNT support both the characterization and improvement of distributed quantum resources. Integrating them is particularly attractive when generating entanglement is costly and measurement access is restricted to peripheral nodes.

Despite their different objectives, entanglement distillation and the QNT protocols considered here share a common operational feature: both rely on local measurements at accessible nodes and classical processing of the outcomes. In distillation, these outcomes determine whether an output pair is retained; in QNT, their statistics are used to infer internal link parameters. This operational similarity raises the possibility of using distillation measurement statistics for network tomography

Casapao et al. pioneered the integration of entanglement distillation and quantum state estimation with Disti-Mator which uses measurement statistics collected during entanglement distillation to estimate the X-, Y-, and Z-basis parameters that fully characterize a Bell-diagonal state under a fidelity condition [6]. Their subsequent work developed a double-selection-based estimator in § II-C using the marginal and joint coincidence statistics collected during distillation [7]; Motivated by these results, we identify a natural starting point for QNT: the states shared by peripheral monitors encode information about the individual links that we use to distribute them. We can therefore not only distill these states but also estimate their parameters, and then use these estimates to infer individual link parameters. We thus integrate QNT into end-to-end distillation by using distillation measurement statistics for link characterization while also generating higher-fidelity entangled pairs, thereby obtaining the best of both worlds. We develop a Joint-Coincidence Estimator (JCE) and a Marginal-Coincidence Estimator (MCE) for this task.

Realizing this idea requires overcoming two challenges. First, state estimation does not automatically imply link identifiability. In a three-link star network with Pauli channels, direct Unicast transmission yields pairwise products of link-level Pauli transfer matrix (PTM) eigenvalues. From these products, one can estimate the absolute value of the link parameters but with a sign ambiguity. To resolve this ambiguity, we adapt the Mergecast and channel-dressing techniques introduced in [8]

to prepare Bell-diagonal states for double-selection measurements. Mergecast provides estimates of three-link correlations, while Clifford dressing makes the required Pauli components accessible. Combining these estimates with those of the pairwise products enables estimation of both the magnitudes and signs of the link parameters without access to measurement outcomes from the central node.

Second, inherited from the double-selection protocol, estimating all three state parameters q_X, q_Y, q_Z requires a source-state fidelity condition, implying another sign ambiguity of the protocol itself. This condition restricts the noise regimes in which JCE applies. We overcome this restriction in MCE by observing that the Z-basis parameter can be uniquely recovered without a fidelity condition. Applying uniform Clifford dressings maps the original X and Y components to the effective Z component. Repeating the estimation in these configurations therefore enables the estimation of all link parameters without the source-state fidelity requirement.

Our contributions are summarized as follows:

- In § IV, we develop a distillation-assisted QNT framework for a three-link star network with unknown Pauli channels. The framework integrates Unicast and adapted Mergecast preparations with double-selection measurements at peripheral monitors, using measurement statistics from both accepted and rejected attempts for estimation. The central node performs prescribed operations without supplying measurement outcomes to the tomography procedure.
- In § IV-D, we construct two estimators to identify the parameters of each Pauli channel in the three-link star network. JCE combines estimates of pairwise and three-link correlations with link-dependent Clifford dressing under the specified source-state fidelity condition $F > 1/2$. MCE combines Z-basis coincidence statistics with uniform Clifford dressing and requires no source-state fidelity condition.
- In § V, we evaluate JCE and MCE through Qiskit simulations and compare their estimation accuracy with existing QNT methods under different measurement budgets and prior assumptions. The results demonstrate the benefit of joint coincidence statistics for estimation and show that both estimators can recover channel parameters while retaining entangled pairs with improved fidelity. We further extend JCE to a six-link triangular network, demonstrating its applicability beyond the three-link star topology. "'

II. Preliminaries

A. Pauli–Liouville Representation

To facilitate the analysis of distillation, we present the Pauli–Liouville representation [9] in the two-qubit scenario below. Let $\mathbb{P}_2 := \{I, X, Y, Z\}^{\otimes 2}$ denote the two-qubit Pauli operator basis, which is a complete orthogonal basis for the operator space $\mathcal{L}\big((\mathbb{C}^2)^{\otimes 2}\big)$ under the Hilbert–Schmidt inner product. Then, every two-qubit operator $\mathcal{G}$ can be uniquely expressed as a linear combination of the elements of $\mathbb{P}_2$. Define the Pauli coefficient associated with $P \in \mathbb{P}_2$ as

$$s_P(\mathcal{G}) := \mathrm{Tr}\big(P^\dagger \mathcal{G}\big)\,, \tag{1}$$

we can express $\mathcal{G}$ as follows,

$$\mathcal{G} = \frac{1}{4}\sum_{P\in\mathbb{P}_2} s_P(\mathcal{G}) P. \tag{2}$$

Following the Pauli-Liouville representation, we introduce the $|\cdot\rangle\!\rangle$ notation, and with it, the state can be expressed as a column vector as follows:

$$|\rho\rangle\!\rangle := \big[s_P(\rho)\big]^T_{P\in\mathbb{P}_2}. \tag{3}$$

For a two-qubit linear map $\mathcal{E}$, its Pauli transfer matrix (PTM) $R_\mathcal{E}$ is a 16×16 matrix, whose enties are

$$(R_\mathcal{E})_{P,Q} := \frac{1}{4}\mathrm{Tr}\big[P^\dagger \mathcal{E}(Q)\big]\,, \text{ for any } P, Q \in \mathbb{P}_2. \tag{4}$$

Consequently, the outcome state of applying the operator $\mathcal{E}$ to an input state ρ can be expressed as follows,

$$|\mathcal{E}(\rho)\rangle\!\rangle = R_\mathcal{E}\,|\rho\rangle\!\rangle. \tag{5}$$

B. Pauli Channels in Bell-Diagonal Basis

Bell basis (composed by Bell states) is particularly convenient for describing two-qubit entanglement under Pauli noise because applying a single-qubit Pauli error to one half of a Bell pair maps it to another Bell state. The four Bell states are

$$|\Phi^\pm\rangle = \frac{|00\rangle \pm |11\rangle}{\sqrt{2}}, \qquad |\Psi^\pm\rangle = \frac{|01\rangle \pm |10\rangle}{\sqrt{2}}. \tag{6}$$

Their density operators admit the Pauli decomposition

$$\begin{aligned}
|\Phi^+\rangle\langle\Phi^+| &= \frac{1}{4}\left(II + XX - YY + ZZ\right),\\
|\Psi^+\rangle\langle\Psi^+| &= \frac{1}{4}\left(II + XX + YY - ZZ\right),\\
|\Psi^-\rangle\langle\Psi^-| &= \frac{1}{4}\left(II - XX - YY - ZZ\right),\\
|\Phi^-\rangle\langle\Phi^-| &= \frac{1}{4}\left(II - XX + YY + ZZ\right).
\end{aligned} \tag{7}$$

These decompositions motivate the definition of the operator subspace

$$\mathcal{S}_\mathrm{B} := \mathrm{span}\{II, XX, YY, ZZ\}. \tag{8}$$

An operator belongs to $\mathcal{S}_\mathrm{B}$ if it is a linear combination of these four Pauli operators.

A two-qubit density operator ρ is called Bell-diagonal if it is diagonal in the Bell basis. By Lemma 1, this is equivalent to $\rho \in \mathcal{S}_\mathrm{B}$.

For such a state, we define its restricted Pauli–Liouville vector as

$$|\rho\rangle\!\rangle_\mathrm{B} := \begin{bmatrix} s_{II}(\rho) & s_{XX}(\rho) & s_{YY}(\rho) & s_{ZZ}(\rho)\end{bmatrix}^T, \tag{9}$$

where $s_P(\rho) := \mathrm{Tr}(P^\dagger\rho)$.

The relation between the Bell basis and the Pauli basis is standard [10, 11]. For completeness of presentation, we state the corresponding change-of-basis relation below.

Lemma 1. *Consider the following ordered bases of $\mathcal{S}_B$:*

$$\begin{aligned}\mathcal{B}_{\text{Bell}} &\coloneqq \left(|\Phi^+\rangle\langle\Phi^+|, |\Psi^+\rangle\langle\Psi^+|, |\Psi^-\rangle\langle\Psi^-|, |\Phi^-\rangle\langle\Phi^-|\right), \\ \mathcal{B}_{\text{P}} &\coloneqq (II, XX, YY, ZZ).\end{aligned} \tag{10}$$

There exists a full-rank transformation matrix W which satisfies $4WW^T = I_4$ such that

$$\mathcal{B}_{\text{Bell}} = \mathcal{B}_{\text{P}} W, \qquad W = \frac{1}{4}\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ -1 & 1 & -1 & 1 \\ 1 & -1 & -1 & 1 \end{bmatrix}. \tag{11}$$

For a normalized state ρ, its fidelity is defined as

$$F(\rho) = \frac{1}{4}\left[s_{II}(\rho) + s_{XX}(\rho) - s_{YY}(\rho) + s_{ZZ}(\rho)\right]. \tag{12}$$

Consider the single-qubit Pauli channel

$$\mathcal{P}(\sigma) = A\sigma + BX\sigma X + CY\sigma Y + DZ\sigma Z, \tag{13}$$

where $A, B, C, D \geq 0$ and $A + B + C + D = 1$. If the first qubit is unaffected and the second qubit undergoes $\mathcal{P}$, the corresponding two-qubit channel is $\mathcal{E} = \mathcal{I} \otimes \mathcal{P}$, where $\mathcal{I}$ denotes the identity channel. Applying this channel to the maximally entangled state $|\Phi^+\rangle$ produces the Bell-diagonal state

$$\begin{aligned}\rho &= \mathcal{E}\left(|\Phi^+\rangle\langle\Phi^+|\right) \\ &= A\,|\Phi^+\rangle\langle\Phi^+| + B\,|\Psi^+\rangle\langle\Psi^+| + C\,|\Psi^-\rangle\langle\Psi^-| + D\,|\Phi^-\rangle\langle\Phi^-|.\end{aligned} \tag{14}$$

As defined in (4), the full PTM $R_{\mathcal{E}}$ is a 16×16 matrix with respect to the complete two-qubit Pauli basis. However, $\mathcal{S}_{\text{B}}$ is invariant under $\mathcal{E}$, meaning that $\mathcal{E}(\mathcal{G}) \in \mathcal{S}_{\text{B}}$ for every $\mathcal{G} \in \mathcal{S}_{\text{B}}$. Therefore, the action of $\mathcal{E}$ on Bell-diagonal states is completely described by its restriction to $\mathcal{S}_B$. With respect to the ordered basis $\mathcal{B}_{\text{P}}$ in 10, the resulting 4×4 PTM is

$$\begin{aligned} R_{\mathcal{E}}|_{\mathcal{S}_B} &= \text{diag}(1, q_X, q_Y, q_Z), \\ q_X &= A + B - C - D = 2(A + B) - 1, \\ q_Y &= A - B + C - D = 2(A + C) - 1, \\ q_Z &= A - B - C + D = 2(A + D) - 1. \end{aligned} \tag{15}$$

Here, q_X, q_Y, q_Z are also the eigenvalues of the single-qubit Pauli channel on X, Y, Z, respectively. It is obvious that $q_X, q_Y, q_Z \in [-1, 1]$, which leads to the identification challenge in QNT, we will elaborate this later in §IV-A.

Consequently, the output state in (14) has the Pauli decomposition

$$\rho = \frac{1}{4}\left(II + q_X XX - q_Y YY + q_Z ZZ\right), \tag{16}$$

and its restricted Pauli–Liouville vector is

$$|\rho\rangle\rangle_{\text{B}} = \begin{bmatrix} 1 & q_X & -q_Y & q_Z \end{bmatrix}^T. \tag{17}$$

By Lemma 1, the Pauli coefficients of state ρ uniquely determine the channel parameters A, B, C, D.

C. Double-Selection Protocol

Entanglement distillation aims to probabilistically transform multiple copies of a noisy entangled pairs into fewer pairs with higher fidelity to a target Bell state, which we take to be $|\Phi^+\rangle$. For example, the protocol considered here uses two noisy ancillary pairs to probabilistically increase the fidelity of a third pair. Recent work has shown that measurement statistics collected during entanglement distillation can also be used for state estimation [7].

We consider the double-selection protocol introduced by Fujii and Yamamoto [12]. Figure 1 illustrates its dual role: successful attempts yield purified entangled pairs, while measurement records from all attempts are used to estimate the input Bell-diagonal state under the conditions specified below.

Alice and Bob initially share three independent copies of the Bell-diagonal state (called *source state*) in (16) through a noisy quantum channel. Denote the copies by ρ_1, ρ_2, ρ_3, with $\rho_1 = \rho_2 = \rho_3 = \rho$.

The protocol proceeds as follows.

1) Alice and Bob each locally apply a CNOT gate, using their qubit from ρ_1 as the control and their qubit from ρ_2 as the target.
2) They each locally apply a second local CNOT gate, using their qubit from ρ_3 as the control and their qubit from ρ_2 as the target.
3) They measure ρ_2 in the Z basis and ρ_3 in the X basis. Let z_A, z_B and x_A, x_B denote the local outcomes of Alice and Bob, respectively.
4) They retain ρ_1 if and only if $z_A = z_B$ and $x_A = x_B$; otherwise, they discard it.

We repeat the double-selection circuit and comparing Alice's and Bob's measurement outcomes yields the marginal and joint coincidence probabilities

$$\begin{aligned} p_Z &\coloneqq \Pr(z_A = z_B) = \frac{1 + q_Z^3}{2}, \\ p_X &\coloneqq \Pr(x_A = x_B) = \frac{1 + q_X^2}{2}, \\ p_{XZ} &\coloneqq \Pr(z_A = z_B,\ x_A = x_B) = \frac{1 + q_Z^3 + q_X^2 + q_Z q_Y^2}{4}. \end{aligned} \tag{18}$$

Here, p_{XZ} is also the *acceptance probability* of the protocol.

The following estimation result follows from the double-selection statistics derived by Casapao et al. [7, Sec. III-A].

Lemma 2 (Estimation from double-selection statistics)**.** *Let ρ be the Bell-diagonal state in* (16)*. If $F(\rho) > 1/2$, then, under ideal local operations and measurements, the coincidence probabilities p_X, p_Z, p_{XZ} uniquely determine the nontrivial PTM eigenvalues q_X, q_Y, q_Z. Specially, from rearranging* (18)*, we have*

$$\begin{aligned} q_Z &= \sqrt[3]{2p_Z - 1}, \\ q_X &= \sqrt{2p_X - 1}, \\ q_Y &= \sqrt{\frac{4p_{XZ} - 2p_X - 2p_Z + 1}{\sqrt[3]{2p_Z - 1}}}. \end{aligned} \tag{19}$$

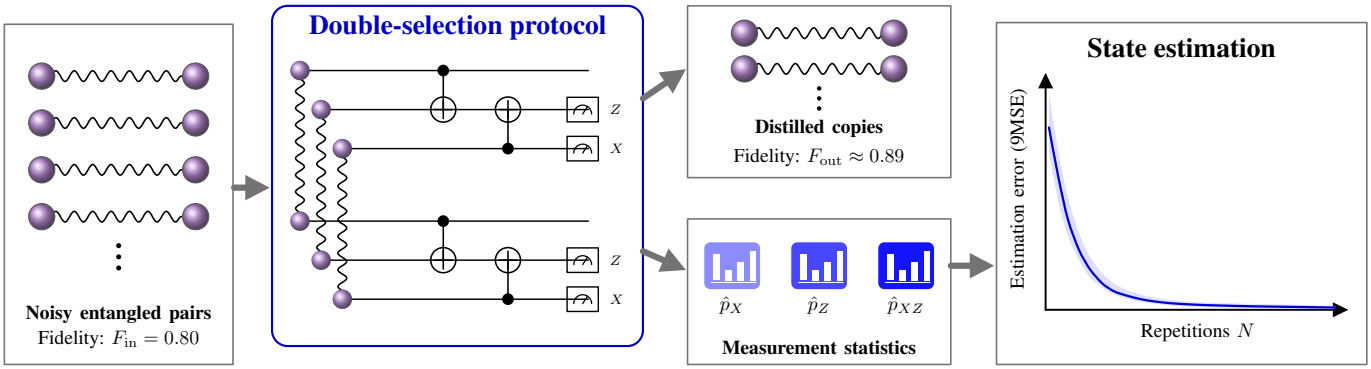


Fig. 1: Illustration of the double-selection protocol, with an example demonstrating its dual role in entanglement distillation and state estimation. Initially, Alice and Bob share independent copies of a Bell-diagonal state with fidelity 0.80 with respect to $|\Phi^+\rangle$ and equal weights on the other three Bell states. The subsystems A_i and B_i form the ith shared entangled pair. In each attempt, three pairs undergo bilateral CNOT operations, after which the second and third pairs are measured in the Z and X bases, respectively. Under ideal local operations and measurements, the first pair is retained when Alice's and Bob's outcomes coincide in both bases, yielding an output state with fidelity approximately 0.89. Meanwhile, measurement records from all attempts, including unsuccessful ones, are used to estimate the input Bell-diagonal state.

Together with (17) *and Lemma 1, these quantities uniquely determine the input Bell-diagonal state.*

Observation 3. *The estimation of parameter q_Z in the double-selection protocol is unique by itself and does not rely on the fidelity condition $F(\rho) > \frac{1}{2}$.*

Later on, we will leverage this observation and another channel dressing technique to remove the requirement of the $F(\rho) > \frac{1}{2}$ condition in our protocols.

III. Problem Formulation

Star Quantum Network. To ease the presentation, let us first consider a three-link star quantum network, illustrated in Fig. 2, is represented by a graph $\mathcal{G} := (\mathcal{N}, \mathcal{L})$, where $\mathcal{N}$ denotes the set of nodes and $\mathcal{L}$ denotes the set of quantum links connecting them. We model each quantum link as a Pauli channel. Note that more general quantum channels can be converted into effective Pauli channels through Pauli twirling [13, 14].

Here, C is the central node while N_1, N_2, N_3 are peripheral nodes, as illustrated in Fig. 2. Each peripheral node N_ℓ is connected to the central node C through one quantum link which is described by a Pauli channel with unknown channel parameters,

$$\mathcal{P}_\ell(\sigma)=A_\ell\sigma+B_\ell X\sigma X+C_\ell Y\sigma Y+D_\ell Z\sigma Z, \ell\in\{1,2,3\}, \quad (20)$$

where parameters $A_\ell, B_\ell, C_\ell, D_\ell \geq 0$ and $A_\ell + B_\ell + C_\ell + D_\ell = 1$. Its nontrivial PTM eigenvalues $q_{X,\ell}, q_{Y,\ell}, q_{Z,\ell}$ are defined in (15). Each link is therefore completely characterized by either $(A_\ell, B_\ell, C_\ell, D_\ell)$ or $(q_{X,\ell}, q_{Y,\ell}, q_{Z,\ell})$. We impose the following nonzero condition on the link channels to simplify the estimation analysis by excluding degenerate cases[1].

Assumption 4 (Nonzero PTM eigenvalues)**.** *For every link $\ell \in \{1,2,3\}$, all nontrivial PTM eigenvalues are nonzero:*

$$q_{P,\ell} \neq 0, \qquad P \in \{X,Y,Z\}, \quad \ell \in \{1,2,3\}. \quad (21)$$

Peripheral Monitors and End-to-End Access. The peripheral nodes N_1, N_2, N_3, referred to as *monitors* , provide the state preparation and measurement (SPAM) capabilities used for tomography. The central node C can route quantum systems, perform prescribed local operations, including quantum gates S, H, and controlled-NOT (CNOT) gates, and discard auxiliary qubits. However, its SPAM capabilities, if present, are inaccessible to the tomography procedure.

We emphasis that this access model follows the end-to-end monitoring framework in the classical network tomography literature and is also adopted in quantum network tomography [8, 18]. It also reflects practical scenarios in which peripheral nodes are accessible to users, whereas the central node is managed by a quantum network service provider that permits prescribed operations but does not grant users access to its internal SPAM capabilities or measurement records.

End-to-End Distillation. Under the access constraints described above, we consider only end-to-end (path-level) entanglement distillation between monitors. The local measurements required at both endpoints by distillation protocols

[1]This condition is sufficient for the proposed estimation but is not necessary for identifiability. If $q_{X,\ell} = q_{Y,\ell} = q_{Z,\ell} = 0$, then $\mathcal{P}_\ell$ is completely depolarizing and $(\mathcal{I} \otimes \mathcal{P}_\ell)(\rho) = I \otimes I/4$ for every Bell-diagonal state ρ [15]. Such a link erases all correlations and can be treated as a failed link using existing link-failure identification methods [16, 17]. If only some PTM eigenvalues vanish, the Clifford dressings in §IV-C can permute the zero and nonzero entries. Alternative configurations may then recover the remaining parameters whenever sufficiently many nonsingular product equations are available. A complete analysis of these degenerate cases is beyond the scope of this paper.

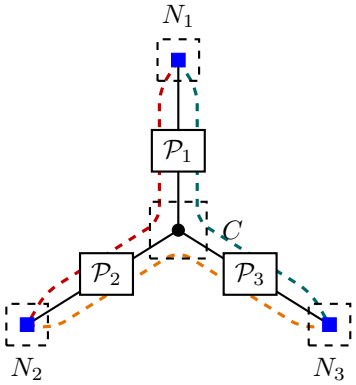


Fig. 2: Three-link star network with peripheral monitors N_1, N_2, N_3 (blue squares) and a central node C (black circle). Each link ℓ is modeled by an unknown Pauli channel $\mathcal{P}_\ell$. The monitors provide SPAM capabilities, while C performs prescribed operations without supplying measurement outcomes to the tomography procedure. The red, teal, and orange dashed lines represent the three end-to-end paths 0 $N_1 - C - N_2$, $N_1 - C - N_3$, and $N_2 - C - N_3$, respectively.

can then be performed at the monitors, with their outcomes accessible to the tomography procedure. Link-level distillation would instead involve measurements at C, whose outcomes are unavailable to users. Even if the service provider performs such distillation internally, we exclude it from our model and use only measurement data obtained from path-level distillation. Specifically, we apply the double-selection protocol described in § II-C to entangled pairs shared between monitors. Besides conditionally retaining an output pair, the protocol provides measurement statistics for estimating the input Bell-diagonal state.

We use these statistics in two estimation schemes. Under the fidelity condition $F(\rho) > 1/2$, Lemma 2 allows all three nontrivial parameters to be estimated using (19). Alternatively, Observation 3 allows the Z-basis parameters to be estimated from p_Z alone without relying on this fidelity condition. Suitable Clifford dressings extend the latter approach to the X and Y components. In both schemes, the estimated end-to-end correlations are then combined to infer the individual link-level channel parameters.

The following assumption ensures the applicability of Lemma 2 to JCE. Note that it is not required by MCE, which is based on Observation 3.

Assumption 5 (Fidelity condition for JCE)**.** *The Bell-diagonal states ρ_{12}, ρ_{13}, $\rho_{m,23}$, and $\rho_{d,23}$, defined in* (23)*,* (24)*, and* (26)*, each have fidelity greater than $1/2$ with respect to $|\Phi^+\rangle$.*

Objective. Given the three-link star network $\mathcal{G}$, our objective is to design end-to-end protocols that use distillation measurement outcomes to estimate all link-level channel parameters $q_{P,\ell}$, for $\ell \in \{1,2,3\}$ and $P \in \{X,Y,Z\}$. We quantify the estimation accuracy using the mean square error (MSE),

$$\epsilon_{\text{est}} := \frac{1}{9}\sum_{\ell=1}^{3}\sum_{P\in\{X,Y,Z\}} \left(\widehat{q}_{P,\ell} - q_{P,\ell}\right)^2, \tag{22}$$

where $\widehat{q}_{P,\ell}$ denotes the estimate of $q_{P,\ell}$. Our objective is to minimize this estimation error for a given measurement budget.

Remark 6 (Extension to general network topologies)**.** Due to space limitations, this work focuses on the three-link star topology, which provides a fundamental setting in which individual link parameters can be estimated from end-to-end observations without access to the central node's measurement outcomes.

The proposed framework can be naturally extended to more general quantum-network topologies without degree-two nodes based on two observations. First, any quantum network without degree-two nodes can be viewed as being composed of three-link star substructures. Second, for tomography purposes, two consecutive Pauli channels can be regarded as a single effective Pauli channel. We illustrate this extension through a larger-network example in § V-B. A more comprehensive treatment is omitted due to space limitations.

IV. Distillation-Assisted Network Tomography

In this section, we develop a distillation-assisted tomography protocol for the three-link star network. §IV-A first introduces direct pairwise distillation and explains its identifiability limitation. §IV-B then presents a modified Mergecast construction that provides estimation involving all three links. §IV-C uses Clifford dressing to obtain the additional estimation needed for complete estimation. Finally, § IV-D combines these measurement statistics to construct JCE and MCE and establish the identifiability of the three Pauli channels. JCE requires the specified source-state fidelity condition $F > 1/2$, whereas MCE removes this requirement.

A. Distillation-Assisted Unicast

We begin with direct Unicast [19] transmission between peripheral monitors, where Unicast refers to single-path probe between one sender monitor and one receiver monitor (in the peripheral). The three-link star network contains three two-link paths, each for one pair of monitors as in Fig. 2. Consider the path $N_1 - C - N_2$ shown in Fig. 3. The Unicast procedure is as follows: monitor N_1 prepares a Bell pair $|\Phi^+\rangle$, retains one qubit locally, and sends the other to N_2 through the Pauli channel $\mathcal{P}_1$, the central node C, and the Pauli channel $\mathcal{P}_2$. The resulting state shared by monitor N_1 and N_2 is Bell diagonal, as shown in Fig. 3.

Applying the same procedure along the all three paths, including the other two $N_2 - C - N_3$ and $N_1 - C - N_3$, gives the quantum states with restricted Pauli–Liouville representations as follows,

$$\begin{aligned}
|\rho_{12}\rangle\rangle_{\text{B}} &= \begin{bmatrix} 1 & q_{X,1}q_{X,2} & -q_{Y,1}q_{Y,2} & q_{Z,1}q_{Z,2}\end{bmatrix}^T,\\
|\rho_{13}\rangle\rangle_{\text{B}} &= \begin{bmatrix} 1 & q_{X,1}q_{X,3} & -q_{Y,1}q_{Y,3} & q_{Z,1}q_{Z,3}\end{bmatrix}^T,\\
|\rho_{23}\rangle\rangle_{\text{B}} &= \begin{bmatrix} 1 & q_{X,2}q_{X,3} & -q_{Y,2}q_{Y,3} & q_{Z,2}q_{Z,3}\end{bmatrix}^T.
\end{aligned} \tag{23}$$

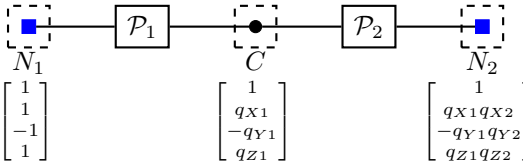


Fig. 3: Two-link path $N_1{-}C{-}N_2$ through the central node C. The column vectors below the nodes are the restricted Pauli–Liouville representation of the shared state at the corresponding transmission stage. By symmetry, the same protocol can be applied to the other two two-link paths, $N_1 - C - N_3$ and $N_2 - C - N_3$.

For each path, the two monitors independently generate three copies of the shared state via the Unicast transmission above and then perform the double-selection protocol in §II-C.

Whenever the source state for distillation in (23) satisfies the fidelity condition of Lemma 2, the double-selection statistics in (18) *uniquely estimate* its nontrivial Pauli correlations (e.g., the last three entries of $|\rho_{12}\rangle\rangle_{\mathrm{B}}$: $\widehat{q_{X,1}q_{X,2}}$, $\widehat{q_{Y,1}q_{Y,2}}$ and $\widehat{q_{Z,1}q_{Z,2}}$). Thus, applying this procedure to all three paths estimate any $\widehat{q_{P,\ell}q_{P,\ell'}}$ for any $P \in \{X,Y,Z\}$ and $(\ell,\ell') \in \{(1,2),(1,3),(2,3)\}$.

a) Identifiability challenge: For each $P \in \{X,Y,Z\}$, the double-selection statistics yield estimates of the pairwise products $q_{P,1}q_{P,2}$, $q_{P,1}q_{P,3}$, and $q_{P,2}q_{P,3}$, each involving an even number of factors. From these three estimates, one can, for instance, estimate the parameter $q_{P,1}$ as $\pm\sqrt{\frac{\widehat{q_{P,1}q_{P,2}}\times\widehat{q_{P,1}q_{P,3}}}{\widehat{q_{P,2}q_{P,3}}}}$, with a sign ambiguity. That is, there are two possible sign choices for each Pauli component, across all three components giving up to eight choices in total. This ambiguity suggests that this protocol alone cannot uniquely identify all channels.

To resolve this sign ambiguity, we introduce a Mergecast protocol in the following subsection, which generates estimates involving all three links (i.e., products of an odd number of factors). By combining the estimates with the ones in Unicast, we resolve this sign ambiguity and uniquely estimate the channel parameters.

B. Distillation-Assisted Mergecast

Fig. 4 illustrates the Mergecast protocol [8], which introduces correlations involving all three links to resolve the sign ambiguity. We apply this protocol to the path $N_2{-}C{-}N_3$ and then use the double-selection protocol to estimate the resulting Bell-diagonal state. The procedure is as follows.

1) Node N_2 prepares $|\Phi^+\rangle$, retains one qubit, and sends the other to C through $\mathcal{P}_2$. Meanwhile, N_1 sends an auxiliary qubit in $|0\rangle$ to C through $\mathcal{P}_1$.
2) Node C applies a CNOT with the auxiliary qubit as the control and the Bell-pair qubit as the target. It then traces out (i.e., discard) the control and forwards the target to N_3 through $\mathcal{P}_3$.
3) After generating three independent copies from repeating steps 1) and 2), node N_2 and N_3 perform double-selection protocol.

Before distillation, each source state shared by N_2 and N_3 is

$$|\rho_{\mathrm{m},23}\rangle\rangle_{\mathrm{B}} = \begin{bmatrix}1 & q_{X,2}q_{X,3} & -q_{Z,1}q_{Y,2}q_{Y,3} & q_{Z,1}q_{Z,2}q_{Z,3}\end{bmatrix}^T. \tag{24}$$

By Lemma 2, the double-selection statistics yield estimates of the nontrivial Pauli coefficients of $\rho_{\mathrm{m},23}$. For example, combining the resulting estimate of $q_{Z,1}q_{Z,2}q_{Z,3}$ with the previously obtained estimate of $q_{Z,1}q_{Z,2}$ gives $\widehat{q}_{Z,3} := \frac{\widehat{q_{Z,1}q_{Z,2}q_{Z,3}}}{\widehat{q_{Z,1}q_{Z,2}}}$, This illustrates how combining the estimates from the two protocols resolves the sign ambiguity. Further details are presented in § IV-D.

To recover the remaining parameters $q_{X,\ell}$ and $q_{Y,\ell}$, we need to access correlations in other Pauli bases. We achieve this through channel dressing, as described below.

Remark 7 (Fidelity inefficiency of the Mergecast protocol)**.** Although this Mergecast operation may reduce the fidelity of the final source state in (24) shared by node N_2 and N_3, it provides information that cannot be obtained from Unicast and resolve the sign ambiguity issue. In other words, it trades some end-to-end fidelity for the identifiability.

C. Clifford-Dressed Pauli Channels

We adapt the channel-dressing technique of Wang et al. [8] to the Mergecast protocol. Specifically, in Fig. 4, we apply different Clifford dressings to $\mathcal{P}_2$ and $\mathcal{P}_3$, while leaving $\mathcal{P}_1$ unchanged. Dressing the channels is equivalent to change input and output bases of the channels, providing the statistics for estimating other channel parameters.

The dressing uses the Hadamard gate H and phase gate S. Both are Clifford gates, which bijectively map Pauli operators to Pauli operators up to a sign under conjugation [20]. Ignoring signs, H exchanges X and Z, while S exchanges X and Y. Together, these exchanges generate all permutations of $\{X,Y,Z\}$. When a Pauli channel is dressed by a Clifford gate and its Hermitian conjugate, the signs cancel, so the dressing simply permutes its PTM eigenvalues. We use the Clifford gates SHS and S to define

$$\mathcal{P}_2^{(SHS)} := SHS \circ \mathcal{P}_2 \circ (SHS)^\dagger,\ \mathcal{P}_3^{(S)} := S \circ \mathcal{P}_3 \circ S^\dagger. \tag{25}$$

These dressings exchange the Y and Z eigenvalues of $\mathcal{P}_2$, and the X and Y eigenvalues of $\mathcal{P}_3$, respectively, as illustrated in Fig. 5.

Replacing $\mathcal{P}_2$ and $\mathcal{P}_3$ with these dressed channels in the Mergecast-based distillation protocol produces

$$|\rho_{\mathrm{d},23}\rangle\rangle_{\mathrm{B}} = \begin{bmatrix}1 & q_{X,2}q_{Y,3} & -q_{Z,1}q_{Z,2}q_{X,3} & q_{Z,1}q_{Y,2}q_{Z,3}\end{bmatrix}^{\mathsf{T}}. \tag{26}$$

By Lemma 2, the double-selection statistics recovers the nontrivial Pauli coefficients of $\rho_{\mathrm{d},23}$. In particular, the last two coefficients provide the information needed to determine $q_{X,3}$ and $q_{Y,2}$. Together with the direct and undressed Mergecast measurements, they enable the complete channel estimation described next.

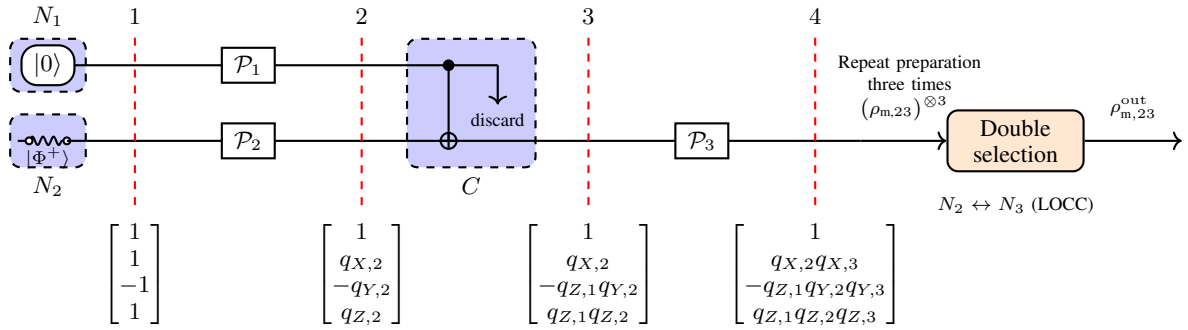


Fig. 4: Mergecast preparation followed by double-selection protocol. Slices 1–4 illustrate the entanglement-generation procedure, adapted from the Mergecast protocol in [8]. The column vectors give the restricted Pauli–Liouville representations of the reduced two-qubit state. The preparation is repeated independently three times, after which monitors N_2 and N_3 perform double-selection protocol to distill a retained entangled pair and estimate the input-state parameters from the measurement statistics.

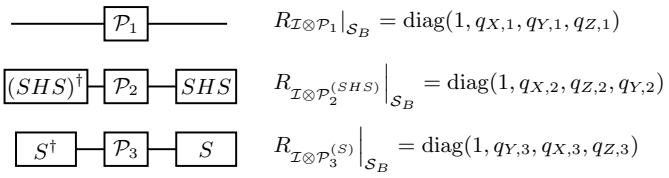


Fig. 5: Link-dependent Clifford dressing used in the modified Mergecast protocol. Each channel $\mathcal{P}_\ell$ is preceded by $U_\ell^\dagger$ and followed by U_ℓ, with $U_1 = I$, $U_2 = SHS$, and $U_3 = S$. Thus, $\mathcal{P}_1$ remains unchanged, while the PTM eigenvalues of $\mathcal{P}_2$ and $\mathcal{P}_3$ are permuted as $(q_{X,2}, q_{Y,2}, q_{Z,2}) \mapsto (q_{X,2}, q_{Z,2}, q_{Y,2})$ and $(q_{X,3}, q_{Y,3}, q_{Z,3}) \mapsto (q_{Y,3}, q_{X,3}, q_{Z,3})$, respectively. These dressings allow the Mergecast protocol to access the mixed products $q_{Z,1}q_{Z,2}q_{X,3}$ and $q_{Z,1}q_{Y,2}q_{Z,3}$, which provide the additional information needed to resolve the remaining sign ambiguities.

D. Link-Parameter Estimation

a) Joint-Coincidence Estimator (JCE): Under Assumption 5 on source-state fidelity, JCE combines the estimates obtained from the four configurations to estimate the individual channel parameters.

By Lemma 2, the double-selection statistics of the directly generated states ρ_{12} and ρ_{13} yield estimators $\widehat{\pi}_{12}^{(P)}$ and $\widehat{\pi}_{13}^{(P)}$ of the pairwise products defined as follows,

$$\pi_{12}^{(P)} \coloneqq q_{P,1}q_{P,2},\ \pi_{13}^{(P)} \coloneqq q_{P,1}q_{P,3},\ P \in \{X, Y, Z\}. \tag{27}$$

Similarly, the double-selection statistics of the undressed Mergecast state $\rho_{\mathrm{m},23}$ yield an estimator $\widehat{\tau}_Z$ of the three-link correlation

$$\tau_Z \coloneqq q_{Z,1}q_{Z,2}q_{Z,3}.$$

Combining these estimates, we define

$$\widehat{q}_{Z,1} \coloneqq \frac{\widehat{\pi}_{12}^{(Z)}\widehat{\pi}_{13}^{(Z)}}{\widehat{\tau}_Z}, \widehat{q}_{Z,2} \coloneqq \frac{\widehat{\tau}_Z}{\widehat{\pi}_{13}^{(Z)}}, \widehat{q}_{Z,3} \coloneqq \frac{\widehat{\tau}_Z}{\widehat{\pi}_{12}^{(Z)}}. \tag{28}$$

To estimate the remaining parameters, we use the double-selection statistics of the dressed Mergecast state $\rho_{\mathrm{d},23}$. These statistics yield estimators $\widehat{\mu}_X$ and $\widehat{\mu}_Y$ of

$$\mu_X \coloneqq q_{Z,1}q_{Z,2}q_{X,3}, \qquad \mu_Y \coloneqq q_{Z,1}q_{Y,2}q_{Z,3}.$$

Dividing these estimates by the corresponding estimates gives

$$\widehat{q}_{X,3} \coloneqq \frac{\widehat{\mu}_X}{\widehat{\pi}_{12}^{(Z)}}, \qquad \widehat{q}_{Y,2} \coloneqq \frac{\widehat{\mu}_Y}{\widehat{\pi}_{13}^{(Z)}}. \tag{29}$$

These expressions provide estimates of both the magnitude and sign of one parameter in each of the X and Y sectors. The remaining parameters are then estimated sequentially as

$$\begin{aligned} \widehat{q}_{X,1} &\coloneqq \frac{\widehat{\pi}_{13}^{(X)}}{\widehat{q}_{X,3}}, \quad \widehat{q}_{X,2} \coloneqq \frac{\widehat{\pi}_{12}^{(X)}}{\widehat{q}_{X,1}}, \\ \widehat{q}_{Y,1} &\coloneqq \frac{\widehat{\pi}_{12}^{(Y)}}{\widehat{q}_{Y,2}}, \quad \widehat{q}_{Y,3} \coloneqq \frac{\widehat{\pi}_{13}^{(Y)}}{\widehat{q}_{Y,1}}. \end{aligned} \tag{30}$$

Finally, applying the transformation in Lemma 1 gives estimators of the Pauli-error probabilities of each channel:

$$\begin{bmatrix} \widehat{A}_\ell \\ \widehat{B}_\ell \\ \widehat{C}_\ell \\ \widehat{D}_\ell \end{bmatrix} \coloneqq W^T \begin{bmatrix} 1 \\ \widehat{q}_{X,\ell} \\ -\widehat{q}_{Y,\ell} \\ \widehat{q}_{Z,\ell} \end{bmatrix}, \qquad \ell \in \{1, 2, 3\}. \tag{31}$$

To summarize, (28)–(30) give estimates of the link parameters, while (31) gives the corresponding estimates of the Pauli-error probabilities. The corresponding exact estimation establishes the following identifiability result.

Theorem 8 (Estimation with a fidelity assumption)**.** *Under Assumptions 4 and 5, all nine link-level PTM eigenvalues, and hence all three Pauli channels, are uniquely identifiable from double-selection statistics through* (28)–(30)*, by combining the Unicast and Mergecast protocols with link-dependent Clifford dressing.*

The fidelity assumption can be relaxed when estimating $q_{Z,1}$, $q_{Z,2}$, and $q_{Z,3}$, as follows.

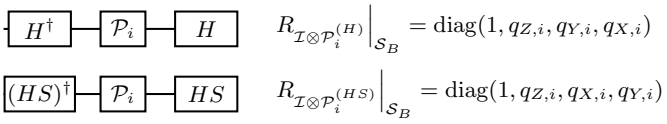


Fig. 6: Clifford dressing of the Pauli channel $\mathcal{P}_i$ and the resulting permutations of its PTM eigenvalues. The upper and lower circuits implement the H and HS dressings, respectively, with the corresponding restricted PTMs shown on the right. These dressings give $q_{Z,i}^{(H)} = q_{X,i}$ and $q_{Z,i}^{(HS)} = q_{Y,i}$, allowing the original X and Y parameters to be estimated through the effective Z parameter.

Observation 9 (Estimation of the link parameters $q_{Z,\ell}$)**.** *The exact Z-basis coincidence probabilities obtained from ρ_{12}, ρ_{13}, and $\rho_{m,23}$ uniquely determine $q_{Z,1}$, $q_{Z,2}$, and $q_{Z,3}$ without Assumption 5. The corresponding estimators are given by* (28)*.*

Indeed, Observation 3 allows the products $q_{Z,1}q_{Z,2}$, $q_{Z,1}q_{Z,3}$, and $q_{Z,1}q_{Z,2}q_{Z,3}$ to be estimated from the corresponding Z-basis measurement statistics alone. These products suffice to estimate all three $q_{Z,\ell}$.

b) Marginal-Coincidence Estimator (MCE): Building on Observation 9, MCE removes the fidelity requirement, constituting another key contribution of this work. We retain (28) for estimating $q_{Z,\ell}$ and extend the same estimation to $q_{X,\ell}$ and $q_{Y,\ell}$ through uniform Clifford dressing. This scheme uses only the Z-basis measurement statistics, leaving the other information unused.

To estimate $q_{X,\ell}$ and $q_{Y,\ell}$, we apply each of the Clifford dressings shown in Fig. 6 uniformly to all three channels in the network. This differs from JCE, which applies different dressings to different channels in Mergecast. The H and HS dressings map $q_{X,\ell}$ and $q_{Y,\ell}$, respectively, to the effective Z eigenvalues of the dressed channels:

$$q_{Z,\ell}^{(H)} = q_{X,\ell}, \qquad q_{Z,\ell}^{(HS)} = q_{Y,\ell}, \qquad \forall \ell \in \{1,2,3\}. \tag{32}$$

For each dressing, we apply the same transformation to all three links and repeat the Unicast and undressed Mergecast procedures with the dressed channels. Observation 3 then yields estimates $\widehat{\pi}_{12}^{(P)}$, $\widehat{\pi}_{13}^{(P)}$, and $\widehat{\tau}_P$ of $q_{P,1}q_{P,2}$, $q_{P,1}q_{P,3}$, and $q_{P,1}q_{P,2}q_{P,3}$, respectively, for $P \in \{X, Y\}$. Substituting these estimates into (28), with Z replaced by X or Y, gives

$$\begin{aligned} \widehat{q}_{X,1} &\coloneqq \frac{\widehat{\pi}_{12}^{(X)}\widehat{\pi}_{13}^{(X)}}{\widehat{\tau}_X}, \quad \widehat{q}_{X,2} \coloneqq \frac{\widehat{\tau}_X}{\widehat{\pi}_{13}^{(X)}}, \quad \widehat{q}_{X,3} \coloneqq \frac{\widehat{\tau}_X}{\widehat{\pi}_{12}^{(X)}}, \\ \widehat{q}_{Y,1} &\coloneqq \frac{\widehat{\pi}_{12}^{(Y)}\widehat{\pi}_{13}^{(Y)}}{\widehat{\tau}_Y}, \quad \widehat{q}_{Y,2} \coloneqq \frac{\widehat{\tau}_Y}{\widehat{\pi}_{13}^{(Y)}}, \quad \widehat{q}_{Y,3} \coloneqq \frac{\widehat{\tau}_Y}{\widehat{\pi}_{12}^{(Y)}}. \end{aligned} \tag{33}$$

Together with (28), these estimators give the following estimation result.

Theorem 10 (Estimation without a fidelity assumption)**.** *Even without Assumption 5, all link-level PTM eigenvalues of these three Pauli channels in the star network remain uniquely identifiable from double-selection statistics via* (28) *and* (33) *by combining Clifford dressing with the Unicast and Mergecast protocols.*

The two estimators establish identifiability under different conditions and exhibit a trade-off between measurement efficiency and applicability. JCE exploits joint X/Z coincidence statistics and link-dependent Clifford dressing to estimate all nine channel parameters using four measurement configurations, but requires the source-state fidelity condition $F > 1/2$. In contrast, MCE relies only on Z-basis coincidence statistics and uniform Clifford dressing, removing the fidelity requirement at the cost of increasing the number of configurations from four to nine. Consequently, each complete round requires 16 measured qubits for JCE and 36 for MCE. Thus, JCE is more resource-efficient in terms of configurations and measurements per round when the fidelity condition holds, whereas MCE offers broader applicability without this condition. The statistical efficiency of the two estimators under equal measurement budgets is further evaluated in § III.

V. Numerical Experiments

We evaluate the proposed joint-coincidence estimator (`JCE`) and marginal-coincidence estimator (`MCE`) through quantum circuit simulations using Qiskit [21]. We first consider a three-link star network and compare the estimation accuracy of the proposed methods with two existing tomography methods: `Mergecast` [XW: cite], which supports general Pauli-channel tomography, and Multicast [22], which is restricted to depolarizing-channel tomography. The comparisons are conducted under different measurement budgets and assumptions about prior channel knowledge. We also examine the number of fidelity-improved entangled pairs retained during distillation to demonstrate the dual functionality of the proposed methods. We then extend `JCE` to a six-link triangular network [XW: (Figure **??**)] to evaluate its applicability beyond the three-link star topology. Estimation accuracy is quantified by the mean squared error (MSE) of the estimated Pauli-channel parameters, with measurement outcomes generated during distillation reused for channel estimation.

A. Performance in a Three-Link Star Network

1) Network setting: We consider a three-link star network in which each link is modeled by a depolarizing channel with $q_{X,\ell} = q_{Y,\ell} = q_{Z,\ell}$. The channel parameters are $q_{P,1} = 0.88$, $q_{P,2} = 0.86$, and $q_{P,3} = 0.84$ for links 1, 2, and 3, respectively.

Local gates and measurements are assumed to be ideal. We consider two estimation settings that differ in whether the depolarizing structure is known beforehand. Each measurement budget is evaluated using 1000 Qiskit trials.

The measurement budget counts all measured qubits, including those measured in rejected distillation attempts, and excludes retained output pairs and unmeasured qubits discarded at the central node. For JCE and MCE, these measurements are part of distillation; for the baseline protocols, they are dedicated to tomography. Thus, "free distillation measurement" denotes reuse without additional tomography measurements, rather than zero physical resource consumption.

TABLE I: Mean numbers of fidelity-improved Bell pairs over 1,000 Qiskit trials at approximately $200,000$ measurements per method. JCE and MCE use independent records, with four and nine configurations per round, respectively. Fidelities describe theoretical input and accepted-output ensembles. Mergecast retains no distilled outputs.

F_{in}	F_{out}	JCE	MCE	Mergecast
0.817600	0.901788	7,618	10,161	0
0.804400	0.891105	7,355	9,800	0
0.748456	0.843859	12,713	8,477	0
Total		**27,687**	**28,438**	**0**

a) Without a depolarizing prior.: Although the simulated channels are depolarizing, the estimators in Fig. 7(a) are only supplied with the general Pauli-channel model. JCE, MCE, and Mergecast therefore independently estimate all nine parameters $\{q_{P,\ell} : P \in \{X,Y,Z\}, \ell = 1,2,3\}$, without imposing equality between the three Pauli components.

A complete round uses four, nine, and nine configurations for JCE, MCE, and Mergecast, respectively, and 16, 36, and 9 measurements. Their performance is compared at equal measurement budgets.

The MSE decreases as the measurement budget increases. Mergecast achieves the lowest MSE in this setting, while JCE is more accurate than MCE at the same measurement budget. The proposed estimators additionally use their measurement records for distillation, producing the retained entangled pairs discussed below.

b) With a depolarizing prior.: Figure 7(b) considers the case in which all methods know the depolarizing structure. Only the three parameters $(q_{P,1}, q_{P,2}, q_{P,3})$ are estimated, using maximum likelihood estimation.

The adapted JCE and MCE both execute three double-selection configurations, ρ_{12}, ρ_{13}, and $\rho_{\mathrm{m},23}$, and measure 12 qubits per round. JCE uses the joint X/Z coincidence statistics, whereas MCE uses the Z-coincidence marginal of the same records. Mergecast uses two Unicast configurations and one Mergecast configuration, measuring three qubits per round. The Multicast baseline [5] measures two qubits per round.

Multicast achieves the lowest MSE, while JCE outperforms Mergecast and MCE. Because JCE and MCE use identical distillation records in this setting, their difference demonstrates the benefit of retaining the joint coincidence information for estimation.

2) Comparison of the two estimators: Fig. 7(c) and Fig. 7(d) compare the X-, Y-, and Z-component MSEs of JCE and MCE, each averaged over the three links. Both estimators reconstruct the channel parameters without imposing the depolarizing constraint.

At equal measurement budgets, JCE achieves lower component MSEs in this setting. At equal round counts, MCE achieves lower MSEs for the X and Y components, while the Z-component MSEs are comparable. For MCE, the different Pauli components are inferred from Z-basis coincidence statistics under Clifford dressing; their errors therefore depend on both the measurement statistics and the subsequent inversion. These comparisons use different resources: each round of MCE executes nine configurations and measures 36 qubits, whereas each round of JCE executes four configurations and measures 16 qubits. Thus, equal round counts do not imply equal measurement budgets. The additional configurations of MCE enable estimation without the source-state fidelity condition. The present experiment compares their accuracy in a regime where both estimators apply.

In addition to estimating channel parameters, both methods retain entangled pairs with improved fidelity. Table I reports the average retained-pair counts at an equal measurement budget. MCE yields a slightly larger total number of fidelity-improved pairs and more pairs in the two higher output-fidelity groups, whereas JCE retains more pairs in the lowest output-fidelity group. The corresponding input and conditional output fidelities are identical for the two methods.

B. Extension to a Six-Link Triangular Network

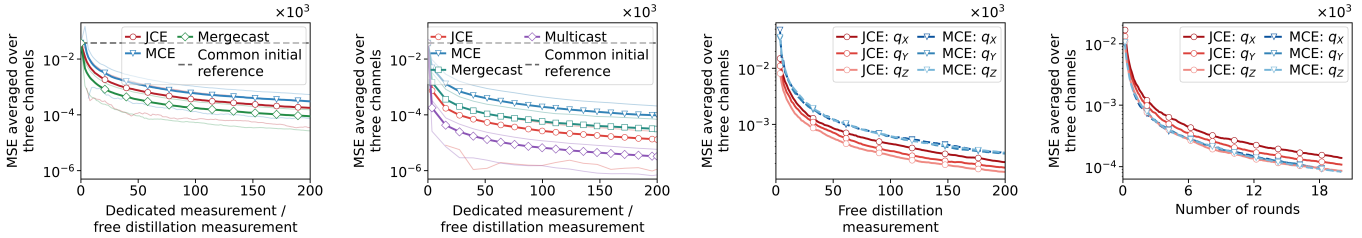
We next consider the network in Fig. 8a, consisting of three internal nodes arranged in a triangle, each connected to one peripheral monitor. In the channel order $(\mathcal{P}_1, \mathcal{P}_2, \mathcal{P}_3, \mathcal{P}_{12}, \mathcal{P}_{13}, \mathcal{P}_{23})$, the true PTM parameters are

$$\boldsymbol{q}_P = \frac{1}{75}(73, 72, 71, 70, 69, 68), \qquad P \in \{X,Y,Z\}. \quad (34)$$

Although the channels are depolarizing, reconstruction estimates all eighteen Pauli-channel parameters independently. We apply JCE to three effective stars centered at C_1, C_2, and C_3. For example, the star centered at C_1 has effective channels $\mathcal{P}_1$, $\mathcal{P}_2 \circ \mathcal{P}_{12}$, and $\mathcal{P}_3 \circ \mathcal{P}_{13}$. For each effective star, we perform two unicast double-selection configurations and two modified Mergecast double-selection configurations, with and without Clifford dressing. In the first stage, we estimate the outer channels and the composite branches. In the second stage, we recover the internal-channel parameters by dividing the estimated composite-branch PTM parameters by the corresponding outer-channel estimates. Consequently, estimation errors in the outer channels also propagate into the internal-channel estimates.

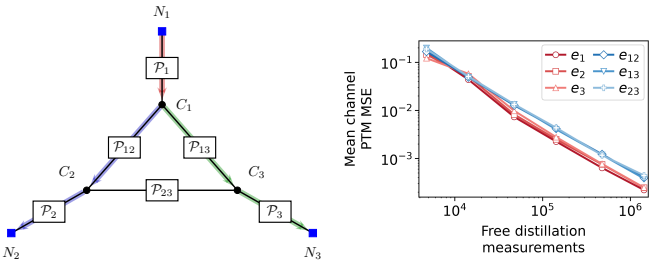
Fig. 8b reports the MSE of each channel, averaged over its three Pauli components and the algebraically valid repetitions. The horizontal axis counts the measurements performed during distillation and reused for tomography, denoted as *free distillation measurements*. One complete round across the three effective stars executes twelve configurations and measures 48 qubits.

The MSE decreases overall for all six channels as the measurement budget increases. In this setting, the outer channels exhibit lower MSE than the internal channels, consistent with the additional error propagation introduced by the second reconstruction stage. These results demonstrate the applicability of the proposed reconstruction procedure to a network beyond the three-link star topology.



(a) Without a depolarizing prior. (b) With a depolarizing prior. (c) Equal round counts. (d) Equal configuration counts.

Fig. 7: Tomography performance in the three-link star network. (a) Comparison of JCE, MCE, and Mergecast at equal measurement counts without a depolarizing prior, using the MSE over nine independent Pauli parameters. (b) Comparison with a known depolarizing model, where all methods estimate three link parameters using maximum likelihood estimation. Light curves in (a) and (b) indicate one standard deviation about the mean; dashed horizontal lines mark the prescribed common zero-data references. (c) Component-wise MSE at equal round counts. Each round executes four configurations for JCE and nine for MCE. (d) Component-wise MSE at equal numbers of configuration executions. With K executions, JCE performs $K/4$ rounds and MCE performs $K/9$ rounds. Component-wise errors are averaged over the three channels. Each budget is evaluated using 1,000 Qiskit trials.



(a) Six-link triangular network. (b) Mean per-channel PTM MSE.

Fig. 8: Network topology and tomography performance in a six-link triangular network. (a) Peripheral monitors N_1, N_2, N_3 connect to internal nodes C_1, C_2, C_3. Colored arrows illustrate an effective star centered at C_1, with channels $\mathcal{P}_1$, $\mathcal{P}_2 \circ \mathcal{P}_{12}$, and $\mathcal{P}_3 \circ \mathcal{P}_{13}$. Analogous constructions at C_2 and C_3 provide the observations used for the two-stage reconstruction. (b) Mean PTM MSE versus the number of distillation measurements reused for tomography. Red curves represent the outer channels e_1, e_2, e_3, and blue curves represent the inner channels e_{12}, e_{13}, e_{23}.

VI. Conclusion

In this work, we presented a distillation-assisted framework for quantum network tomography that uses the same measurement records to characterize internal links and support entanglement distillation. For a three-link star network with unknown Pauli channels, we combined `Unicast` and `Mergecast` preparations with Clifford dressing to develop two estimators that recover link-level channel parameters without additional tomography measurements. The `JCE` exploits marginal and joint coincidence statistics to improve estimation accuracy under a source-state fidelity condition, while the `MCE` removes this requirement through uniform Clifford dressing at the cost of additional configurations. The two estimators therefore offer a trade-off between measurement requirements and applicability.

Qiskit simulations demonstrated that both estimators accurately reconstruct link channels while retaining entangled pairs with improved fidelity in the tested regime. We further extended the framework to a six-link triangular network, demonstrating its applicability beyond the basic star topology. Future work will extend the framework to account for state preparation and measurement (SPAM) errors and explore its application to more general network topologies. We also plan to develop distillation protocols that improve both tomography accuracy and the fidelity of the retained entangled pairs. These results establish distillation-assisted tomography as a promising approach to simultaneous channel characterization and high-fidelity entanglement distribution.

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